



Predictability of complex networks

Fei Jing^{a,b}, Zi-Ke Zhang^{c,d,e,1}, Qingpeng Zhang^{a,b} , and Giorgio Parisi^{d,f,1} 

Contributed by Giorgio Parisi; received December 5, 2025; accepted March 21, 2026; reviewed by Ginestra Bianconi and Tao Zhou

We establish network predictability theory by mapping link prediction onto a spin glass model, where network partitions correspond to spin configurations, and predictability equals the system's average energy. Using the cavity method from statistical physics, we prove that global predictability decomposes into individual link contributions, enabling an efficient local sampling algorithm that reduces the computational complexity of evaluating individual link contributions from being dependent on the entire network to only its local neighborhood, scaling with the average degree. We derive exact results for canonical network models: Erdős–Rényi networks exhibit universal predictability of 0.5 regardless of algorithm choice, establishing the random baseline, while structured networks show predictability controlled by their prior parameters. We introduce the predictability index, which quantifies the maximum achievable performance without information loss and accurately predicts algorithm performance under random division. Analysis of real networks validates our framework, revealing how degree heterogeneity and structural patterns govern predictability. This physics-based approach provides both theoretical insights into link prediction limits and practical tools for assessing network reconstruction potential, with implications for applications from biological network inference to social network analysis.

cavity method | spin glasses | algorithmic performance

The ability to predict missing links, future connections, or evolving structures in complex networks represents one of the most fundamental challenges in network science (1, 2). From predicting protein interactions in biological systems (3) to recommending social connections (4) and anticipating failures in infrastructure networks (5), network predictability touches virtually every domain where complex systems arise. Despite decades of algorithmic development, ranging from similarity-based heuristics to sophisticated machine learning approaches, critical theoretical questions remain unaddressed. What structural features make a network inherently predictable or unpredictable? How does the loss of structural information affect predictability in a quantifiable way? Is there a universal framework for characterizing predictability between different types of networks and prediction tasks?

Recent studies have highlighted the task of establishing theoretical bounds for prediction (6, 7). Early attempts to bridge statistical mechanics and network prediction have tried to quantify predictive limits, such as likelihood analysis (8), perturbation (9) and temporal-topological correlation (10), which optimize Hamiltonians to identify spurious or missing links, serving as a precursor to physics-based predictability.

In addition, previous efforts in statistical physics have demonstrated great power of spin glass models to capture the structure and behaviors of complex systems (11–14). These models, initially developed to understand magnetic materials, have found widespread applications in optimization, machine learning, and complex networks. The key insight is that many complex problems can be mapped onto energy landscapes, where system configurations correspond to different states, and optimal solutions emerge from the principles of energy-minimization (15). In this work, we bridge network science and statistical physics by developing a spin glass framework for network predictability. Our approach transforms the network prediction problem into an average-energy task, where different ways of partitioning the network (for training and testing) correspond to an ensemble of spin configurations, and the prediction performance maps to the system's energy.

The spin formulation yields several theoretical breakthroughs. First, this mapping allows us to apply the cavity method (16) developed for spin glasses: this is a powerful technique from statistical physics used to solve models on sparse graphs or similar contexts (17). By using the cavity method, we show that the global network's predictability decomposes exactly into contributions from individual links. This is not merely an approximation but a fundamental property revealed by the Hamiltonian formulation.

Significance

We establish a theoretical framework for network predictability by mapping the link prediction problem onto a spin glass model from statistical physics. This approach reveals that a network's global predictability decomposes into local contributions, leading to an efficient assessment algorithm and a “Predictability Index” that quantifies maximum achievable performance. Our analysis proves that random networks have a universal predictability of 0.5—the baseline for random guessing—while the predictability of structured networks is determined by their generative parameters. This forges a fundamental link between a network's structure and its inherent reconstructability.

Author affiliations: ^aMusketeers Foundation Institute of Data Science, The University of Hong Kong, Hong Kong 999077, China; ^bDepartment of Pharmacology and Pharmacy, The University of Hong Kong, Hong Kong 999077, China; ^cCenter for Digital Communication Studies, Zhejiang University, Hangzhou 310058, China; ^dInternational Research Center of Complexity Sciences, Hangzhou International Innovation Institute, Beihang University, Hangzhou 311115, China; ^eSchool of Computer, Qinghai Normal University, Xining, Qinghai 810008, China; and ^fDipartimento di Fisica, Sapienza Università di Roma, Roma 00185, Italy

Author contributions: F.J., Z.-K.Z., Q.Z., and G.P. designed research; F.J., Z.-K.Z., Q.Z., and G.P. performed research; Z.-K.Z. analyzed data; and F.J., Q.Z., and G.P. wrote the paper.

Reviewers: G.B., Queen Mary University of London; and T.Z., University of Electronic Science and Technology of China.

The authors declare no competing interest.

Copyright © 2026 the Author(s). Published by PNAS. This article is distributed under [Creative Commons Attribution-NonCommercial-NoDerivatives License 4.0 \(CC BY-NC-ND\)](#).

¹To whom correspondence may be addressed. Email: zkz@zju.edu.cn or giorgio.parisi@roma1.infn.it.

This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2535161123/-DCSupplemental>.

Published April 20, 2026.

This decomposition explains why some networks are inherently more predictable and mathematically justifies our efficient local sampling algorithm. Second, we derive exact results for canonical network models: random graphs exhibit universal predictability of 0.5, while structured networks show predictability controlled by their generative parameters. Third, we introduce the predictability index (PI), which quantifies the maximum achievable prediction accuracy and remains robust even under severe information loss.

Results

Spin Glass Model for Predictability.

Preliminaries. Here, we propose a spin glass based model for the link prediction process, seeking to predict missing connections which are randomly removed from the original network as prediction targets, while the probability of these missing links forming between node pairs is estimated based on a given algorithm, using the remaining links that persist in the network. The performance of the algorithm is quantified by metrics such as the Area Under the Receiver Operating Characteristic Curve (AUC) (18), named by predictability, which measures the effectiveness of distinguishing missing links from nonexistent links. In this paper, we introduce a PI γ_0 that can be computed analytically and is related to AUC (see *SI Appendix, Predictability* for details).

Each existing link $e \in E$ is associated with a spin variable $\sigma_e \in \{1, -1\}$, where $\sigma_e = 1$ represents that e is a missing link; otherwise e is a remaining link (Fig. 1A). The full spin configuration set is denoted as $\sigma = \{\sigma_e\}_{e \in E} \in \{1, -1\}^M$, with $M = |E|$ (Fig. 1 E and F). The system is characterized by a parameter P , representing the fraction of links that are missing. A given algorithm is going to produce a scoring function for links, i.e., $s_e(\sigma)$. It is important to note that this score depends on the given configuration of the network, specifically the set of remaining links defined by σ (Fig. 1 B–D). Furthermore, the analytical decomposition presented later relies on the assumption that $s_e(\sigma)$ is a local scoring function [e.g., common neighbors (CN) (1) and preferential attachment (PA) (19)], where the score of a link is determined exclusively by the structure within nearby links (first or second neighbors depending on the method we follow). We define the algorithm-dependent energy of a spin configuration σ as

$$\mathcal{H}(\sigma) = \sum_{e \in E, \tilde{e} \in \bar{E}} \frac{1 + \sigma_e}{2} h(s_e(\sigma) - s_{\tilde{e}}(\sigma)) + \hat{h} \sum_{e \in E} s_e, \quad [1]$$

where $h(\cdot)$ is the Heaviside step function and $s_e(\sigma)$ is the scoring function for link e for a given prediction algorithm in a particular configuration σ ; the Hamiltonian depends on the scoring function, but we do not indicate this dependence explicitly in order to lighten the notation. Missing links ($\sigma_e = 1$) contribute to score comparisons, while remaining links ($\sigma_e = -1$) only provide scoring information without direct comparison. To ensure the probability of a link being missing is P , the Boltzmann factors must satisfy the ratio as $\frac{e^{-\beta \hat{h}}}{e^{\beta \hat{h}}} = e^{-2\beta \hat{h}} = \frac{P}{1-P}$, which yields that

$$\beta \hat{h} = \frac{1}{2} \ln \frac{1-P}{P}, \quad [2]$$

where β is the inverse temperature parameter. This ensures that in the noninteracting limit, the probability of a link being assigned

$\sigma_e = 1$ is exactly P . The system's thermodynamics is described by the free energy:

$$F = -\frac{1}{\beta} \ln Z, \quad Z = \sum_{\sigma} e^{-\beta \mathcal{H}(\sigma)}. \quad [3]$$

Importantly, we find that the PI (γ) is connected to the spin glass formulation via the normalized average energy, also shown in Fig. 2A:

$$\gamma = \frac{\langle \mathcal{H} \rangle}{M \left(\binom{N}{2} - M \right)} \quad \text{with} \quad \langle \mathcal{H} \rangle = \lim_{\beta \rightarrow 0} \frac{\partial(\beta F)}{\partial \beta}, \quad [4]$$

In most cases, $\binom{N}{2} \gg M$ for real networks as they are usually sparse enough. Therefore, Eq. 4 effectively bridges predictability with spin glass representation (see *SI Appendix, Predictability* for the detailed mathematical derivation). The reason for this relation is that, in the high-temperature limit ($\beta \rightarrow 0$), the system explores all possible link configurations with equal probability, making the average energy equivalent to the sum of successful score comparisons defined by the AUC statistic.

To rigorously validate our framework, we utilized a diverse array of nine local similarity indices, each representing a distinct topological mechanism for link prediction. We began with fundamental metrics such as CN (1), which posits that nodes sharing multiple neighbors are predisposed to connect, and the Jaccard Index (20), which normalizes this overlap by the total number of neighbors. We also included PA (19), a method grounded in the “rich-get-richer” phenomenon where high-degree nodes attract more links.

To capture more complex structural bias, we employed degree-normalized indices: the Hub Promoted Index (HPI) (21) and Hub Depressed Index (HDI) (2), which respectively favor or penalize connections to high-degree nodes, and the Leicht–Holme–Newman Index (LHN-I) (22), which normalizes common neighbors by the product of node degrees. Furthermore, we integrated physically inspired methods like resource allocation (RA) (2), which simulates resource transfer by penalizing high-degree common neighbors, and the Cannistraci–Alanis–Ravasi (CAR) index (3), which enhances the CN score by accounting for the local clustering between shared neighbors. Finally, we utilized the stochastic differential method (SDM) to assess structural perturbations caused by potential edge additions. Mathematical definitions for all indices are provided in the *SI Appendix, Experimental Setup*.

Local decomposition of predictability. By modeling link prediction as a spin glass system, we can leverage powerful analytical tools from statistical physics, particularly the cavity method (16). This approach calculates predictability through the ensemble's average energy in the high-temperature limit, effectively capturing the local interaction patterns that govern link formation in complex networks.

In the high-temperature regime with the constraint $\beta \hat{h} = \frac{1}{2} \ln \left(\frac{1-P}{P} \right)$, the system's free energy decomposes into independent single-link contributions:

$$F = \sum_{e \in E} F_e, \quad \text{where} \quad F_e = -\frac{1}{\beta} \ln Z_e. \quad [5]$$

Each single-link free energy takes the form:

$$F_e = -\frac{1}{\beta} \ln Z_e, \quad [6]$$

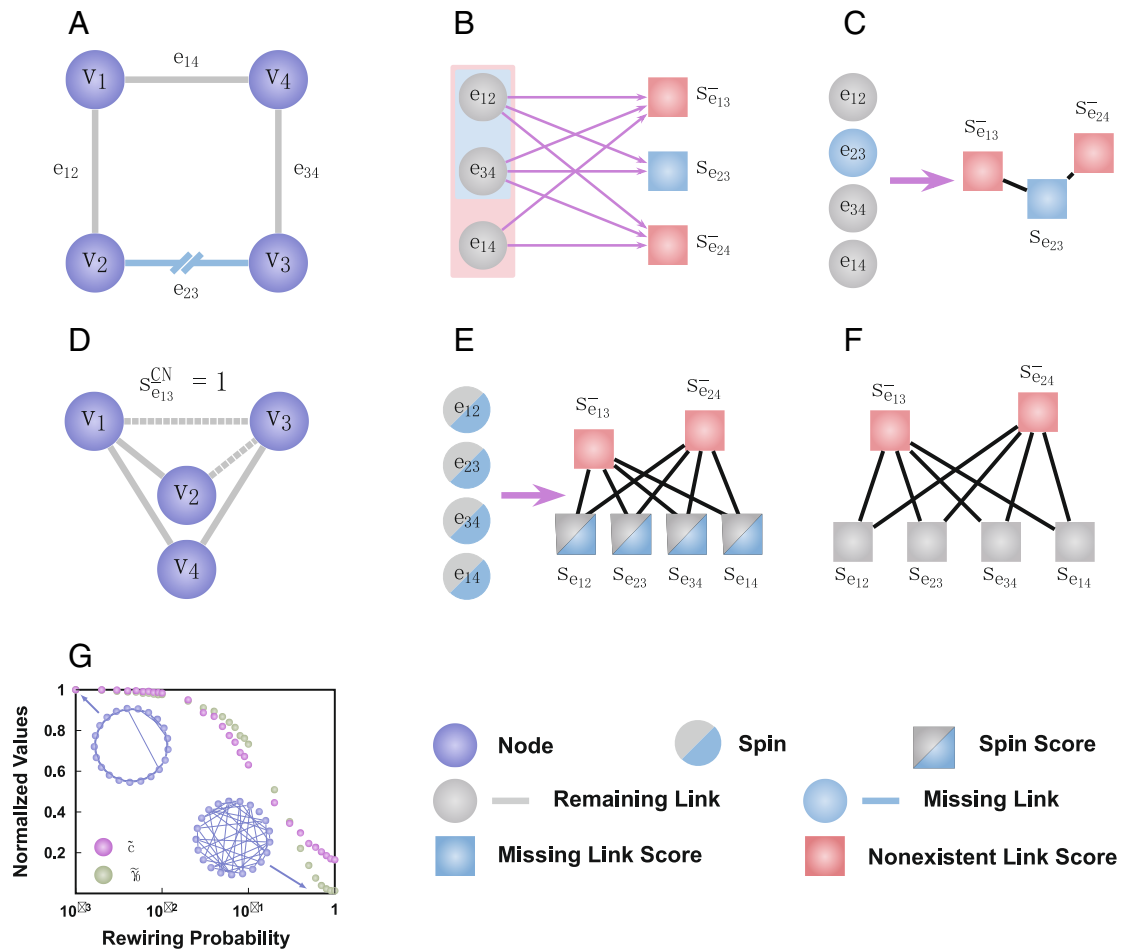


Fig. 1. Spin glass model illustration. (A) Original network with four nodes $V = \{v_1, v_2, v_3, v_4\}$ and four links $E = \{e_{12}, e_{23}, e_{34}, e_{14}\}$, where E is divided into missing links (e_{23}) and remaining links (e_{12}, e_{34}, e_{14}). (B) Scoring link likelihoods from remaining links, in which each link score is dependent on its neighboring spins. For $s_{e_{13}}$ and $s_{e_{24}}$, the neighboring spins are $\{s_{e_{12}}, s_{e_{34}}, s_{e_{14}}\}$; for $s_{e_{23}}$, they are $\{s_{e_{12}}, s_{e_{34}}\}$. (C) Energy calculation via score comparisons. In this configuration, the energy contribution from score comparisons is given by $h(s_{e_{23}} - s_{e_{13}}) + h(s_{e_{23}} - s_{e_{24}})$, leading to the total energy $E(\sigma) = h(s_{e_{23}} - s_{e_{13}}) + h(s_{e_{23}} - s_{e_{24}}) + \hat{h} \sum_{e \in E} \sigma_e$ (Eq. 1). (D) An example of how common neighbor method scores the nonexistent link e_{23} based on its neighboring spins. (E) Ensemble definition through the partition function $Z = \sum_{\{\sigma\} \in \{1, -1\}^4} \exp(-\beta E(\sigma))$. (F) The PI (Eq. 15) provides a direct measure of predictability through score comparison within the original network structure. (G) Normalized clustering coefficient $\tilde{c} = c/c^{\max}$ and predictability $\tilde{\gamma}_0 = (\gamma_0 - \gamma_0^{\min})/(\gamma_0^{\max} - \gamma_0^{\min})$ plotted against rewiring probability in Watts-Strogatz networks.

with the single-link partition function:

$$Z_e = \sum_{\sigma} \exp \left(-\beta \left[\frac{1 + \sigma_e}{2} \sum_{\bar{e} \in \bar{E}} h(s_e(\sigma) - s_{\bar{e}}(\sigma)) + \sum_{e \in E} \hat{h} \sigma_e \right] \right). \quad [7]$$

Here, F_e and Z_e represent the free energy and partition function associated with the local environment of a single link e . This decomposition implies that the global predictability of the network can be understood as an average of local, individual link predictabilities:

$$\gamma = \langle \gamma_e \rangle \text{ with } \gamma_e = \frac{1}{\binom{N}{2} - M} \lim_{\beta \rightarrow 0} \frac{\partial(\beta F_e)}{\partial \beta}, \quad [8]$$

where γ_e is the predictability contribution derived from the local free energy F_e (Fig. 2B). The results under the high-temperature limit correspond to the predictability of any given network under a random division mechanism. The high-temperature limit ensures that every valid partition of the network (respecting the ratio P) is weighted equally, which matches the standard evaluation

protocol in link prediction. Furthermore, the dependence of γ on P is not removed and is not a constant for all links. For every single configuration (representing a specific training/test split), the scores are recalculated based on the particular local topology defined by that split. The thermodynamic calculation then effectively computes the expectation value of these varying scores over the ensemble of all possible random partitions.

This expectation value corresponds precisely to the definition of predictability (AUC). The formula is similar to Eq. 4 with a simplified form for the free energy: F_e at the place of F . Moreover, the fundamental operation of link prediction—removing and reconstructing network edges—directly parallels the spin removal and restoration process central to the cavity method.

Local sampling. The theoretical decomposability of predictability motivates a highly efficient computational strategy, which we term **Local Sampling**. The goal is to estimate predictability without performing computationally expensive simulations over the entire network.

The core insight is that for local prediction algorithms, the score difference between a potential missing link e and a nonexistent link $\bar{e}$ depends only on the state of links in their immediate vicinity, which we define as the **local neighborhood**

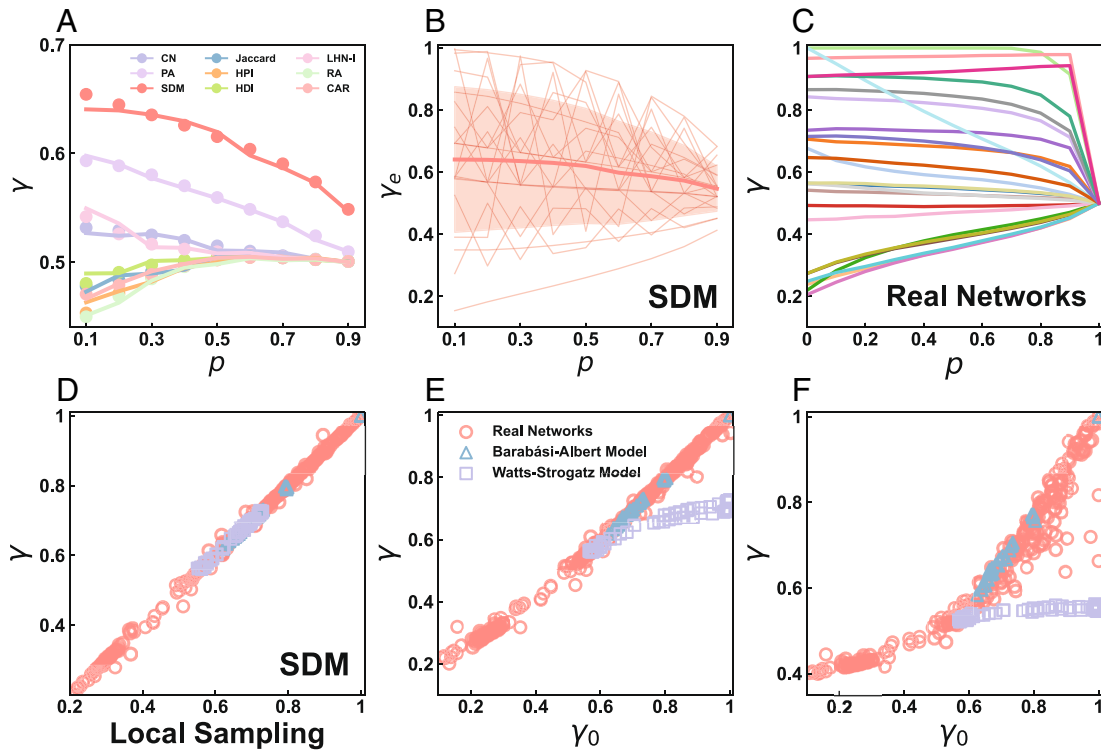


Fig. 2. (A) Average energy γ of the spin glass model and corresponding predictability obtained by nine local-topology-based link prediction methods. Results are evaluated on synthetic networks generated by the Barabási-Albert model. (B) Individual average energy γ_e derived from the cavity field analysis for the SDM. Each curve represents the contribution from a single link within networks also generated by the Barabási-Albert model. The light-colored area represents the SD of the average energy of all individuals (refer to *SI Appendix, Local Sampling* for more details). (C) Predictability for SDM vs. the ratio of missing links on some real networks (see *SI Appendix, Fig. S5* for the full dataset). (D) Local sampling predictability vs. traditional link prediction predictability on Barabási-Albert model, Watts-Strogatz model, and real networks in the dataset. (E and F) PI γ_0 vs. predictability γ for SDM with the ratio of missing (E) links $P = 0.2$ and (F) $P = 0.8$.

$\mathcal{N}_{e,\bar{e}}$. Therefore, instead of simulating the random division of all M links in the network, we can accurately approximate the expected score comparison by considering random divisions only within this local neighborhood, while treating all links outside it as fixed (i.e., remaining).

This procedure is formalized by defining a local Hamiltonian for the pair $(e, \bar{e})$:

$$\mathcal{H}_{e,\bar{e}} = \frac{1 + \sigma_e}{2} b(s_e - s_{\bar{e}}) + \sum_{e' \in \mathcal{N}_{e,\bar{e}}} \hat{h} \sigma_{e'}. \quad [9]$$

The corresponding local partition function is $Z_{e,\bar{e}} = \sum_{\sigma \mathcal{N}_{e,\bar{e}}} \exp(-\beta \mathcal{H}_{e,\bar{e}})$, where the sum runs only over the spin configurations $\sigma \mathcal{N}_{e,\bar{e}}$ of the neighborhood links.

This approach dramatically reduces computational complexity. A full simulation requires partitioning all M links. In contrast, local sampling requires partitioning only $|\mathcal{N}_{e,\bar{e}}|$ links. For algorithms based on local similarity indices, this neighborhood size is typically proportional to the average node degree $\langle k \rangle$, which is much smaller than M . This reduces the complexity of evaluating a single pair's contribution to predictability from an operation dependent on the entire network to one that scales with its local environment. As shown in *Fig. 2D*, this method yields results in excellent agreement with full-network simulations while offering a substantial computational speedup, enabling the analysis of much larger networks.

Structural Features. The random division mechanism fundamentally transforms how we analyze structural features in link prediction. Rather than working with fixed feature values from

the original network, algorithms must contend with random variables arising from the sampling process. This transformation introduces both uncertainty and correlations among features, significantly complicating theoretical analysis.

Consider a structural feature, such as common neighbors. In the original network, each node pair has a fixed number of common neighbors. However, random division converts this deterministic value into a random variable whose distribution depends on the sampling probability P . Moreover, when multiple features depend on overlapping network structures, their random transformations become correlated, adding another layer of complexity.

To analyze these transformations systematically, we employ generating functions. For the common neighbor feature, we define the original distributions as

$$G_{ex}(x) = \sum_{c=0}^N S_{ex}(c) x^c, \quad G_{non}(x) = \sum_{c=0}^N S_{non}(c) x^c, \quad [10]$$

where $S_{ex}(c)$ represents the fraction of existing links having exactly c common neighbors, and $S_{non}(c)$ is the corresponding fraction for nonexistent links. After random division, these distributions transform into:

$$G_{ex}^R(x) = \sum_{c=0}^N S_{ex}^R(c) x^c, \quad G_{non}^R(x) = \sum_{c=0}^N S_{non}^R(c) x^c. \quad [11]$$

The key insight is that a common neighbor survives in the remaining structure only if both its incident edges are retained.

This occurs with probability $(1 - P)^2$. Therefore, the generating functions transform according to

$$G_{ex}^R(x) = G_{ex} (1 - (1 - P)^2 + (1 - P)^2 x) \quad [12]$$

$$G_{non}^R(x) = G_{non} (1 - (1 - P)^2 + (1 - P)^2 x) \quad [13]$$

This elegant relationship encodes how binomial edge sampling affects the common-neighbor distributions. For detailed derivations of degree feature and multifeature distributions, see *SI Appendix, Predictability for Structural Features*. This framework provides a unified approach to understanding how random division affects local structural features, enabling precise predictive calculations across different link prediction methods.

Heaviside Step Function. The Heaviside step function $b(\cdot)$ serves as the core mechanism for converting continuous score differences into binary predictability outcomes. Understanding how random division affects this function is crucial for analyzing predictability across different sampling rates.

For the entire network, we must consider the distributions of features across all existing links E and nonexistent links $\bar{E}$. The predictability becomes

$$\gamma = \mathbb{E}_{e \in E, \bar{e} \in \bar{E}} [\mathbb{E}[b(s_e^R - s_{\bar{e}}^R)]] \quad [14]$$

Unlike the individual-pair case, this network-level expectation generally lacks monotonicity (see *SI Appendix, The Heaviside Step Function Under Random Division* for more details). The complex interplay between feature distributions makes the derivative with respect to p analytically intractable (Fig. 2C). However, the boundary behavior remains well-defined:

- As $p \rightarrow 0$ (complete information), the remaining structure approaches the whole network:

$$\lim_{p \rightarrow 0} \gamma = \gamma_0 = \mathbb{E}_{e \in E, \bar{e} \in \bar{E}} [b(s_e - s_{\bar{e}})], \quad [15]$$

which we define γ_0 as the **PI**.

- As $p \rightarrow 1$ (no information), nearly all existing links are removed:

$$\lim_{p \rightarrow 1} \gamma = 0.5. \quad [16]$$

This universal limit of 0.5 confirms that predictability reduces to random guessing when structural information vanishes, regardless of the initial feature distributions or algorithm choice.

The PI, γ_0 , serves as the theoretical upper bound of link separability. Defined as the limit of standard predictability γ as the fraction of missing information $p \rightarrow 0$, it measures the maximum potential of an algorithm to correctly rank existing links (e) higher than nonexistent ones ($\bar{e}$):

$$\gamma_0 = \mathbb{E}_{e \in E, \bar{e} \in \bar{E}} [b(s_e - s_{\bar{e}})]. \quad [17]$$

Our empirical results validate its utility as both a predictive measure and a characterization tool across different sampling regimes. At low removal regime ($P = 0.2$), Fig. 2E reveals a striking near-linear relationship between PI and actual predictability when only 20% of links are removed. However, at high removal regime ($P = 0.8$), PI maintains strong predictive power (Fig. 2F). More detailed results can be shown in *SI Appendix, Figs. S15–S23*.

Predictability in Synthetic Networks. Synthetic networks provide controlled environments for understanding how network generation mechanisms influence predictability. By systematically varying model parameters, we can establish precise relationships between structural patterns and their predictive signatures, offering insights often impossible to obtain from real networks with unknown evolutionary histories.

The ER model establishes the fundamental baseline for predictability analysis. In ER networks, the complete structural symmetry between existing and nonexistent links yields a universal result: for any ER network and any link prediction algorithm, predictability equals exactly 0.5. This invariance (demonstrated in Fig. 3C) arises because all node pairs are statistically equivalent—no algorithm can distinguish between links that exist by chance and those that do not. Consequently, predictability values above 0.5 signal the presence of nonrandom structural patterns amenable to prediction. The complete proof appears in *SI Appendix, Predictability in Synthetic Networks*.

The Barabási–Albert (BA) model introduces PA, creating degree heterogeneity that enables prediction. Using the SDM algorithm as a representative example, Fig. 3A reveals how the power parameter α controls predictability. High α strengthens rich-get-richer dynamics, concentrating existing links around high-degree hubs; while low α distributes connections more uniformly, reducing the distinction between link types. This concentration effect creates clear topological signatures: missing links cluster in high-degree regions, while nonexistent links predominate in the periphery, enhancing algorithmic discrimination.

In the Watts–Strogatz (WS) model, low rewiring probability (P_{rewire}) creates high local clustering (Fig. 1F). When a link is designated as “missing,” it is removed from the observed graph and cannot contribute to the similarity scores (e.g., common neighbors) for itself or other links. This systematically disadvantages missing links compared to nonexistent links that are surrounded by a fully intact local structure, leading to a separation in scores and thus high predictability. As P_{rewire} increases, this local structure is destroyed, and predictability decays toward the random baseline of 0.5 (Fig. 3B).

Our null model analysis (Fig. 3D–F) reveals the fundamental role of degree correlations. Average degree preservations (Fig. 3D) are very similar to ER model, degree sequence preservations (Fig. 3E) show moderate consistency, and joint degree sequence preservation (Fig. 3F) yields invariant predictability. This hierarchy confirms that SDM and similar degree-based methods fundamentally depend on joint degree patterns. The scoring function $s_{xy} = f(k_x, k_y)$ remains unchanged when joint degree sequences are preserved.

Configuration models with controlled degree distributions (Fig. 3G–I) reveal how distribution parameters govern predictability through their effect on heterogeneity. For power-law distribution (Fig. 3G), a smaller exponent parameter produces more heavy-tailed distributions with extreme degree variations, and amplifies the contrast between core and peripheral nodes, substantially increasing predictability. For the exponential distribution (Fig. 3H), a larger scale parameter creates longer tails and exhibits significantly higher predictability. For a normal distribution (Fig. 3I), both smaller means and larger variances amplify the network’s heterogeneity and enhance its predictability.

Conclusions and Discussion

We have established a rigorous theoretical framework for network predictability by bridging link prediction and statistical physics.

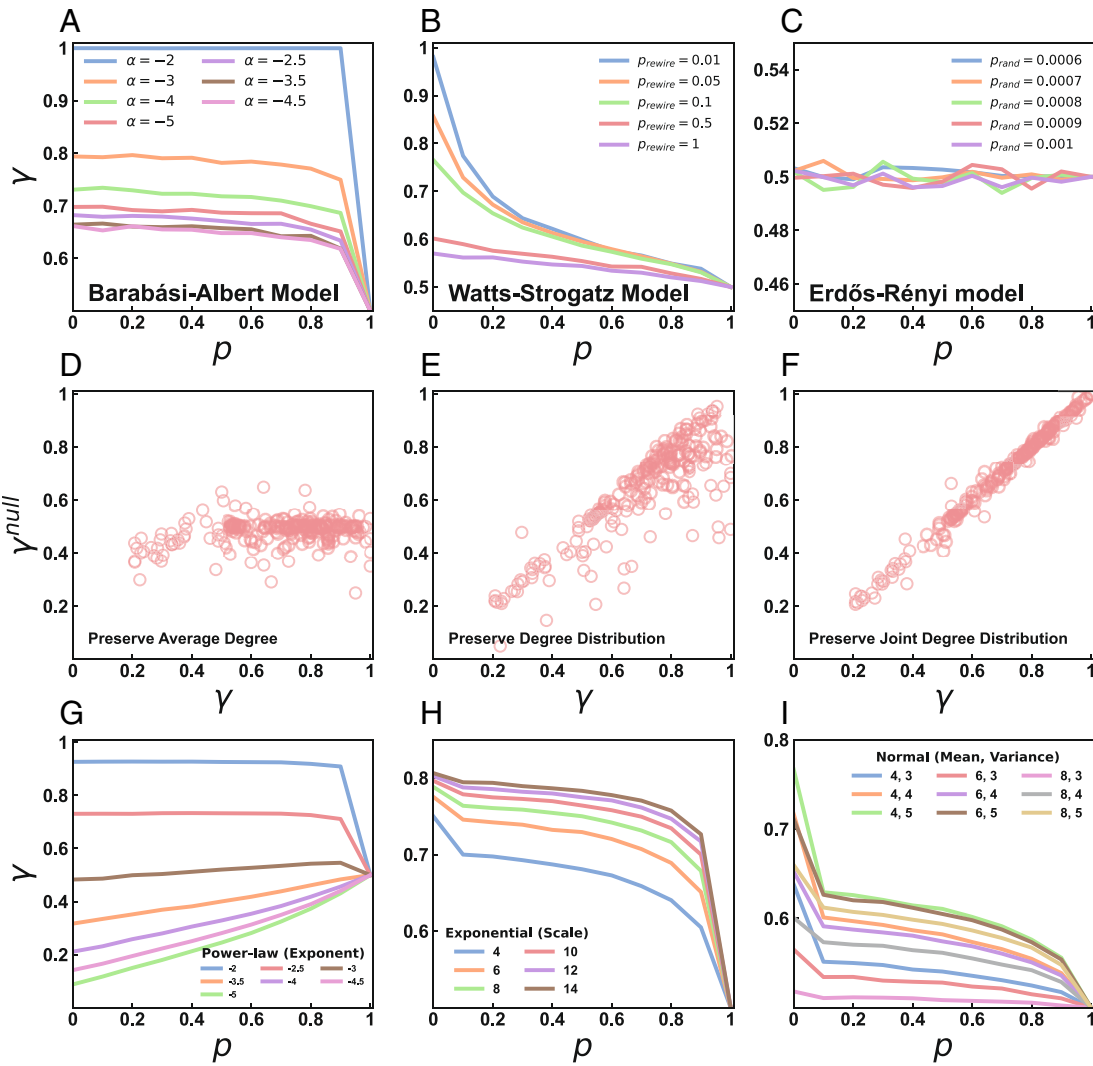


Fig. 3. Influence of prior parameters on network predictability. (A) Power exponent parameter (α) effects in the Barabási-Albert model. (B) Rewiring parameter (P_{rewire}) variation in the Watts-Strogatz model. (C) Connection probability (P_{rand}) in the Erdős-Rényi (ER) model. (D–F) Comparison of predictability between null models derived from real networks and their original counterparts: (D) null model maintaining average degree, (E) null model maintaining degree distribution, and (F) null model maintaining joint degree distribution. (G–I) Predictability analysis in configuration models featuring: (G) power-law degree distribution $P(k) \sim k^\alpha$ with power exponent parameter (α), (H) exponential degree distribution $P(k) \sim e^{-k/\lambda}$ with scale parameter λ , and (I) normal degree distribution $P(k) \sim \mathcal{N}(\mu, \sigma^2)$ with mean (μ) and variance parameters (σ).

Through the mapping of network partitioning onto spin configurations, we transform the abstract concept of predictability into the concrete quantity of average energy, enabling powerful analytical tools from physics to illuminate fundamental limits of network reconstruction. Global predictability emerges as the sum of individual link contributions, revealed through cavity method analysis. This decomposition not only provides theoretical clarity but also enables the computationally efficient local sampling algorithm. We establish how predictability transitions from its maximum value (γ_0) to the random baseline of 0.5 as edge removal increases. This transition curve quantifies the robustness of different algorithms and network structures to incomplete information. The ER result—predictability equals 0.5 for any algorithm—provides a fundamental benchmark. Deviations from this baseline directly measure the exploitable structure within networks. The PI emerges as a powerful diagnostic tool for network analysis. By computing PI before extensive experiments, researchers can assess whether sufficient structural signal exists for meaningful prediction, compare different feature

sets' effectiveness, and estimate required training data sizes for desired performance. While our framework provides a rigorous physics-based interpretation of predictability, two key limitations should be noted. First, the derived predictability γ and the index γ_0 are algorithm-dependent. Unlike information-theoretic entropy, which provides a universal bound based solely on the adjacency matrix, our measure reflects the interplay between the network structure and the specific algorithm used to observe it. Second, the premise that the local sampling process can efficiently perform link prediction is the assumption of local interactions. The efficiency of the single-link decomposition (Eq. 8) holds only for scoring functions like CN, PA, or SDM, which calculate probabilities within a restricted, finite neighborhood. Our framework opens several avenues for future research. The current analysis focuses on specific algorithms. Developing universal upper bounds on predictability—independent of algorithmic choice—would establish fundamental limits

analogous to channel capacity in information theory. Extending the spin glass formulation to global methods [network dynamics (23), matrix factorization (24) and machine learning (25)] requires new theoretical tools to handle their nonlocal scoring functions. Real networks are often dynamic in their evolution or involve diverse relationship types, with recent compelling research revealing how these attributes shape the development of collective intelligence (26, 27). Generalizing our framework to these richer structures could reveal how temporal correlations and layer coupling affect the fundamental limits of predictability. The relationship between predictability and generative mechanisms suggests deeper connections to network evolution theory. High predictability indicates strong organizing principles—forging this connection is a key step toward a unified theory of network structure, evolution, and predictability.

Data, Materials, and Software Availability. All codes and data used to generate the results in this article are publicly available on GitHub at (28). The real-world network datasets analyzed are available for download from the ICON database (29). All other data are included in the manuscript and/or *SI Appendix*.

ACKNOWLEDGMENTS. This work was supported by the National Natural Science Foundation of China (Grant No. 72371224), the Kunlun Talent Program of Qinghai Province (Grant No. [2024]1), the National Foundation for Social Sciences (Key Project, Grant No. 21AXW006), the Major Project of Philosophy and Social Sciences Research of Ministry of Education (Grant No. 2025JZDZ008), the General Research Fund (Grant No. 17209225) and the Strategic Topic Fund (Grant No. STG2/P-705/24-R) of the Research Grants Council of Hong Kong, the Seed Fund for Collaborative Research of The University of Hong Kong (Grant No. 2407102490), the Fundamental Research Funds for the Central Universities, Zeng Xianzi Funds for the International Exchanges and ZJU-USYD Ignition Grants.

1. D. Liben-Nowell, J. Kleinberg, "The link prediction problem for social networks" in *Proceedings of the Twelfth International Conference on Information and Knowledge Management* (2003), pp. 556–559.
2. T. Zhou, L. Lü, Y. C. Zhang, Predicting missing links via local information. *Eur. Phys. J. B* **71**, 623–630 (2009).
3. C. V. Cannistraci, G. Alanis-Lobato, T. Ravasi, From link-prediction in brain connectomes and protein interactomes to the local-community-paradigm in complex networks. *Sci. Rep.* **3**, 1613 (2013).
4. L. Backstrom, J. Leskovec, "Supervised random walks: Predicting and recommending links in social networks" in *Proceedings of 4th ACM International Conference on Web Search and Data Mining* (2011), pp. 635–644.
5. D. Zhang, J. Yin, X. Zhu, C. Zhang, Network representation learning: A survey. *IEEE Trans. Big Data* **6**, 3–28 (2018).
6. T. Zhou, Progresses and challenges in link prediction. *iScience* **24**, 103217 (2021).
7. E. Xu et al., Predictability of complex systems. *Phys. Rep.* **1166**, 1–107 (2026).
8. L. Pan, T. Zhou, L. Lü, C. K. Hu, Predicting missing links and identifying spurious links via likelihood analysis. *Sci. Rep.* **6**, 22955 (2016).
9. L. Lü, L. Pan, T. Zhou, Y. C. Zhang, H. E. Stanley, Toward link predictability of complex networks. *Proc. Natl. Acad. Sci. U.S.A.* **112**, 2325–2330 (2015).
10. D. Tang et al., Predictability of real temporal networks. *Natl. Sci. Rev.* **7**, 929–937 (2020).
11. J. Reichardt, S. Bornholdt, Statistical mechanics of community detection. *Phys. Rev. E* **74**, 016110 (2006).
12. M. Mézard, A. Montanari, *Information, Physics, and Computation* (Oxford University Press, 2009).
13. L. Zdeborová, F. Krzakala, Statistical physics of inference: Thresholds and algorithms. *Adv. Phys.* **65**, 453–552 (2016).
14. H. Zhou, Long-range frustration in a spin-glass model of the vertex-cover problem. *Phys. Rev. Lett.* **94**, 217203 (2005).
15. F. Jing, Z. K. Zhang, Y. C. Zhang, Q. Zhang, Data organization limits the predictability of binary classification. *arXiv [Preprint]* (2024). <https://doi.org/10.48550/arXiv.2401.17036> (Accessed 12 July 2024).
16. M. Mézard, G. Parisi, M. A. Virasoro, *Spin Glass Theory and Beyond: An Introduction to the Replica Method and Its Applications* (World Scientific Publishing Company, 1987), vol. 9.
17. Z. Liu, Y. Zheng, *AI Ethics and Governance* (Springer, 2022).
18. J. Hanley, B. McNeil, The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology* **143**, 29–36 (1982).
19. A. L. Barabási, R. Albert, Emergence of scaling in random networks. *Science* **286**, 509–512 (1999).
20. P. Jaccard, Distribution de la flore alpine dans le bassin des dranses et dans quelques régions voisines. *Bull. Soc. Vaudoise Sci. Nat.* **37**, 241–272 (1901).
21. E. Ravasz, A. L. Somera, D. A. Mongru, Z. Oltvai, A. L. Barabási, Hierarchical organization of modularity in metabolic networks. *Science* **297**, 1551–1555 (2002).
22. S. Leitch, *Prosperity for All in the Global Economy-World Class Skills* (The Stationery Office, 2006).
23. T. Carletti, L. Giambagli, G. Bianconi, Global topological synchronization on simplicial and cell complexes. *Phys. Rev. Lett.* **130**, 187401 (2023).
24. A. K. Menon, C. Elkan, "Link prediction via matrix factorization" in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases* (Springer, 2011), pp. 437–452.
25. A. Ghasemian, H. Hosseinmardi, A. Galstyan, E. M. Airolidi, A. Clauset, Stacking models for nearly optimal link prediction in complex networks. *Proc. Natl. Acad. Sci. U.S.A.* **117**, 23393–23400 (2020).
26. Y. Meng, S. P. Cornelius, Y. Y. Liu, A. Li, Dynamics of collective cooperation under personalised strategy updates. *Nat. Commun.* **15**, 3125 (2024).
27. Y. Meng, A. McAvoy, A. Li, Promoting collective cooperation through temporal interactions. *Proc. Natl. Acad. Sci. U.S.A.* **122**, e2509575122 (2025).
28. F. Jing, Theoretical Link Prediction. GitHub. https://github.com/Feijing92/theoretical_link_prediction. Deposited 2 April 2026.
29. A. Clauset, E. Tucker, M. Sainz, The Colorado Index of Complex Networks. ICON. <https://icon.colorado.edu>. Deposited 30 May 2016.